

CP1a cubic-SOS common-parent compatibility certificate

Verification-first research certificate - T0 scope

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- Authority: T0 kinematic/static common-parent compatibility certificate only
- Claim-bearing: no

0. Result first

There is a single finite-regulator real-scalar Hamiltonian family on one periodic three-torus whose critical quadratic Hessian contains the exact three-mode PA-H1 frequency fixture and whose negative-control branch has an exact eight-node finite-wavevector classical ordering window. Both roles use the same field, Weyl algebra, volume, boundary condition, $\hbar$, regulator, Hamiltonian family, state-selection rule, and relative-energy subtraction.

This is a genuine repair of the decoupled-product and cross-model energy-zero defects recorded by EXP-000707, but only at **CP1a** scope. It is not full CP1. The PA-H1 match occurs at $r=0$, whereas ordering requires $r<0$; no dynamical law for $r(\tau)$ or total work ledger has yet been derived. The selected interacting ground state's PA-H1 restriction is not the earlier pure Gaussian state. The fitted node cone is anisotropic and the fourth-order ultraviolet group speed is unbounded. These are physical-selection liabilities, not details to hide.

The energy statement is deliberately narrow:

At fixed finite cutoff, volume, periodic boundary, $\hbar=1$, coupling, and Hamiltonian family, the common subtraction $C_N=E_N(0)$ gives $E_N(r)-E_N(0)<0$ for $r<0$. This compares members of one parameter family. It is not an energy-below-empty-space result and, without CP2, it does not account for the work required to change r .

1. Common finite-regulator object

Let

$$\begin{aligned} L &= \pi/2, \\ T_L^3 &= (R/LZ)^3, \\ \Gamma_N &= \{4n : n \in Z^3, ||n||_\infty \leq N\}, \quad N \geq 1. \end{aligned}$$

Let E_N be the real Fourier space supported on Γ_N , with the usual reality condition. On T^*E_N use canonical coordinates (ϕ, p) , the canonical symplectic form, periodic boundary conditions, inertia one, and $\hbar=1$. The Weyl algebra is generated by this finite-dimensional canonical phase space and is represented on the infinite-dimensional Hilbert space $L^2(E_N)$; neither the operator algebra nor its Hilbert-space representation is finite-dimensional.

For $g>0$, define

$$\begin{aligned} H(N, r) &= 1/2 ||p||_2^2 + F_r[\phi], \\ F_r[\phi] &= 1/2 \langle \phi, (a_0(-i \text{grad}) + r)\phi \rangle \\ &\quad + g/4 \int_{T_L^3} \phi(x)^4 dx. \end{aligned}$$

The symbol is

$$\begin{aligned} a_0(k) &= \alpha [\sum_i (k_i^2 - 16)]^2 \\ &\quad + \beta \sum_{i<j} (k_i^2 - k_j^2)^2, \\ \alpha &= 1/256, \\ \beta &= 21/512. \end{aligned}$$

Equivalently,

$$a_0(k) = \frac{1}{256} \left\{ \sum_i (k_i^2 - 16)^2 + (21/2) \sum_{i < j} (k_i^2 - k_j^2)^2 \right\}.$$

This is a local fourth-order, sign-flip- and permutation-invariant operator. It is a candidate interpolation, not a derived microscopic law.

The finite-dimensional Schrodinger realization is

$$\text{mathsf{H}}_-(N, r) = -1/2 \Delta_{-(E_N)} + F_r.$$

Since

$$\int \phi^4 \geq \|\phi\|_{L^4}^4/V, \\ V=L^3,$$

its potential is confining at every finite N . The standard finite-dimensional positivity-improving Schrodinger theorem therefore gives a unique strictly positive normalized ground state $\Psi_-(N, r)$ and energy $E_-(r)$. This theorem is imported standard analysis; it is not presented as a new TECT result.

2. Coefficients are derived inside the declared ansatz

Consider the two-term cubic sum-of-squares family

$$a_-(\alpha, \beta)(k) = \alpha \sum_i (k_i^2 - 16)^2 + \beta \sum_{i < j} (k_i^2 - k_j^2)^2.$$

The requested finite PA-H1 squared frequencies are inserted calibration data:

$$a(0)=9, \\ a(4e_1)=25.$$

They give the two independent equations

$$9 \cdot 4^4 \cdot \alpha = 9, \\ 4 \cdot 4^4 \cdot \alpha + 2 \cdot 4^4 \cdot \beta = 25.$$

Hence

$$\alpha=1/256, \\ \beta=21/512, \\ \beta/\alpha=21/2.$$

The solution is unique **only within this two-parameter ansatz**. Two coefficients were fixed by two calibration values, so there is no holdout prediction here. Expanding gives the independently checkable local symbol

$$a_0(k)=\frac{1}{256} \left[\begin{aligned} &22 \sum_i k_i^4 \\ &-19 \sum_{i < j} k_i^2 k_j^2 \\ &-96 |k|^2 \\ &+2304 \end{aligned} \right].$$

3. Exact zero set and lattice gap

On the Fourier lattice $k=4n$,

$$a_0(4n) = (\sum_i n_i^2 - 3)^2 + (21/2) \sum_{i < j} (n_i^2 - n_j^2)^2. \quad (3.1)$$

Both terms are nonnegative. Equality forces all three integer squares to be equal and their sum to be three, hence

$$n_i^2=1 \text{ for every } i.$$

The zero set is therefore exactly

$$K_8=\{4(s_1, s_2, s_3): s_i \in \{-1, +1\}\}.$$

There is also an exact off-node gap. Write $x_i=n_i^2$. If the three integer values are not all equal, at least two pairwise differences are nonzero integers, so

$$\sum_{i < j} (x_i - x_j)^2 \geq 2$$

and the anisotropy term alone is at least 21. If they are equal, write $x_i = m^2$. Excluding the node value $m=1$, (3.1) becomes

$$9(m^2-1)^2 \geq 9,$$

with equality at $m=0$. Consequently

$$\begin{aligned} \Delta_{\text{lattice}} &= \min \{a_0(4n) : 4n \text{ not in } K_8\} \\ &= 9. \end{aligned} \quad (3.2)$$

This proof is global on $\mathbb{Z}^3$; the executable finite-cube scan is only a regression check.

4. Node geometry and an early physical warning

At the positive node $Q=(4, 4, 4)$, direct differentiation gives

$$\begin{aligned} D^2 a_0(Q) &= \begin{bmatrix} 11 & -19/4 & -19/4 \\ -19/4 & 11 & -19/4 \\ -19/4 & -19/4 & 11 \end{bmatrix}. \end{aligned}$$

At other sign choices the off-diagonal entries are conjugated by the sign matrix. Every node therefore has spectrum $\{3/2, 63/4, 63/4\}$.

The radial eigenvector is the node sign vector and the two transverse eigenvalues are $63/4$. With unit inertia, the squared-speed ratio is

$$(63/4)/(3/2)=21/2.$$

Thus the exact fitted node cone is strongly anisotropic. Moreover, on the large positive first axis,

$$\begin{aligned} \omega(K, 0, 0) &\sim (\sqrt{22}/16) K^2, \\ d\omega/dK &\sim (\sqrt{22}/8) K. \end{aligned}$$

The ultraviolet group speed is unbounded. CP1a is therefore a useful compatibility witness but is not presently a survivor of a common isotropic Lorentz-cone or causal-UV test.

5. Exact PA-H1 quadratic calibration

The circumference of the existing PA-H1 finite fixture is also $L=\pi/2$. Embed a one-dimensional normalized real mode by

$$(Uf)(x_1, x_2, x_3) = f(x_1)/L.$$

The two transverse integrals contribute L^2 , so U is isometric. It sends the existing normalized basis to the normalized T3 modes

$$\begin{aligned} &L^{-3/2}, \\ &\sqrt{2}L^{-3/2} \cos(4x_1), \\ &\sqrt{2}L^{-3/2} \sin(4x_1). \end{aligned}$$

At $r=0$, their potential eigenvalues are

$$\begin{aligned} a_0(0) &= 9, \\ a_0(\pm 4e_1) &= 25. \end{aligned}$$

Together with unit kinetic coefficients, the phase-space quadratic form pulls back to

$$\text{diag}(9, 25, 25, 1, 1, 1),$$

and the canonical symplectic form pulls back unchanged. The frequencies are therefore exactly 3, 5, 5, so the already registered PA-H1 characteristic map applies to this **quadratic three-mode calibration**.

This does not derive the full Klein--Gordon dispersion. For example the next transverse-constant mode does not satisfy the KG value. Nor is the three-mode space invariant under the unchanged local cubic force:

$$(a+b \cos 4x)^3$$

contains the second harmonic with coefficient $3ab^2/2$, and $\cos^3(4x)$ contains the third harmonic with coefficient $1/4$. A minimal $N=1$ Galerkin projection hides these modes, but that closure is a regulator artifact and fails along the nested cutoff sequence.

6. Exact failure of the unchanged componentwise PA-M2 kernel

The current componentwise PA-M2 quadratic symbol has the form

$$a_{\text{cmp}}(k) = c \sum_i (k_i^2 - 16)^2.$$

It has the desired eight zeros, but a single scale cannot also match both PA-H1 calibration values. Requiring $a_{\text{cmp}}(0) = 9$ fixes

$$c = 3/256.$$

Then

$$a_{\text{cmp}}(4e_1) = 2 \cdot c \cdot 4^4 = 6,$$

not 25. This is a sharp same-torus, same-node, single-scale obstruction. It does not exclude higher invariants, auxiliary fields, nonlocal kernels, or a different calibration target. The CP1a mixed cubic invariant is the minimal repair inside the declared two-SOS family, not evidence that nature uses it.

7. One fixed parameter does not realize both roles

Adding r shifts the calibration squared frequencies to

$$\begin{aligned} 9+r, \\ 25+r. \end{aligned}$$

Even after a common time rescaling, preservation of the PA-H1 ratio requires

$$(25+r)/(9+r) = 25/9,$$

which is equivalent to $r=0$. By contrast, the finite-wavevector classical ordering result below requires $r < 0$.

Thus one fixed Hamiltonian in the family does not simultaneously give the exact PA-H1 calibration and an ordered PA-M2 branch. A nonstationary control law and its energy exchange are not optional bookkeeping; they are the CP2 gate.

8. Finite-regulator classical ordering theorem

Theorem CP1a-static

For every $N \geq 1$, $g > 0$, and

$$-9 \leq r < 0,$$

the finite-dimensional functional F_r has a global minimizer, every global minimizer has a nonzero K_8 component and is inhomogeneous, and every nonzero stationary point obeys

$$\|\phi\|_2^2/V \leq |r|/g, \quad (8.1)$$

$$\begin{aligned} \|\mathcal{P}_{K_8}^\perp \phi\|_2^2 / \|\phi\|_2^2 \\ \leq |r|/9. \end{aligned} \quad (8.2)$$

Proof

Coercivity and finite dimension give existence. Choose any Q in K_8 and the real trial

$$\phi(x) = A \cos(Q \cdot x).$$

Using the exact torus averages $\int \cos^2 = V/2$ and $\int \cos^4 = 3V/8$,

$$F_r[\phi]/V = r A^2/4 + 3g A^4/32.$$

The nonzero optimum is

$$\begin{aligned} A^2 &= -4r/(3g), \\ F_r[\phi]/V &= -r^2/(6g) < 0. \end{aligned} \quad (8.3)$$

On the orthogonal complement of K_8 , (3.2) and $r \geq -9$ imply

$$\langle \phi, (a_0 + r)\phi \rangle \geq 0.$$

The quartic term is also nonnegative. Every configuration with zero node component therefore has energy at least zero, whereas (8.3) is negative. Every global minimizer must have a node component and hence is inhomogeneous.

For a nonzero stationary point, testing the Euler--Lagrange equation against ϕ gives

$$\langle \phi, a_0 \phi \rangle + r \|\phi\|_2^2 + g \int \phi^4 = 0.$$

Since $a_0 \geq 0$ and $\int |\phi|^4 = \int |\phi|^2^2/V$, (8.1) follows. Also

$$\begin{aligned} 9 \int |P_{\perp}(K_8 \text{ perpendicular}) \phi|^2 & \\ & \leq \langle \phi, a_0 \phi \rangle \\ & \leq \int r |\phi|^2, \end{aligned}$$

which is (8.2). This completes the proof.

The node set is the same CI8 set used by PA-M2, so its exact kernel-restricted quartic convolution identity remains available as a leading node-space comparison. The theorem above does not assert that the full nonlinear minimizer lies exactly in that non-invariant subspace or that one morphology is physically selected.

9. This is not a decoupled product

Let $e_0 = V^{1/2}$ and let $e_Q = (2/V)^{1/2} \cos(Q \cdot x)$ be normalized. The coefficient of $a^2 b^2$ in

$$g/4 \int (a e_0 + b e_Q)^4$$

is

$$(g/4)^6 \int e_0^2 e_Q^2 = 3g/(2V) > 0.$$

The calibration and ordering modes therefore interact in one local quartic Hamiltonian. This directly separates CP1a from the vacuous direct-product control in the strict-interface no-go.

10. Selected state and common relative-energy ledger

Use the same rule at every r : select the unique finite-cutoff interacting ground state $\Psi_i(N, r)$. Its restriction to the PA-H1 calibration Weyl subalgebra is a well-defined state, so one algebra and one state-selection rule now carry both roles.

The restriction is not the old pure quasi-free PA-H1 state. If a subsystem marginal of a pure state were exactly pure, the full state would factorize across that subsystem. For any participating oscillator,

$$\begin{aligned} x^4 |0\rangle & \\ & = [3|0\rangle + 6 \sqrt{2}|2\rangle + 2 \sqrt{6}|4\rangle]/(4 \omega^2). \end{aligned}$$

Because $\int h^4 > 0$, the local quartic interaction creates a nonzero fourth-excitation coefficient

$$g \left(\int h^4 \right) \sqrt{6}/(8 \omega^2).$$

No quadratic term or term of lower degree in that oscillator cancels this component. The factorized free Gaussian cannot be an interacting eigenstate. The common-parent state repair is therefore an interacting, generally non-Gaussian restriction, not an exact recovery of the earlier Gaussian fixture.

For the energy ledger, define one cutoff-dependent additive subtraction for the whole family:

$$\begin{aligned} C_N &= E_N(0), \\ \text{mathsf{H}}_N(r)^{\text{rel}} &= \text{mathsf{H}}_N(r) - C_N. \end{aligned}$$

Using $\Psi_i(N, 0)$ as a trial state at $r < 0$,

$$\begin{aligned} E_N(r) & \\ & \leq E_N(0) + (r/2) \langle \Psi_i(N, 0), |\phi|^2 \Psi_i(N, 0) \rangle \\ & < E_N(0). \end{aligned}$$

The last expectation is strictly positive. This makes the sign meaningful inside one regulated Hamiltonian family. It does not compare with physical empty space, remove C_N , or include the external work that changes r .

11. Devil's-advocate review

- Objection: the coefficients predict the PA-H1 spectrum.** VALID WITH MITIGATION. They were fit to 9 and 25; the manifest labels them inserted calibration targets, and there is no holdout prediction.
- Objection: uniqueness selects this kernel.** VALID WITH MITIGATION. Uniqueness holds only in the declared two-SOS cubic family. Other local invariants, more fields, nonlocal kernels, and different regulators remain open.
- Objection: the same physical state realizes PA-H1 and order.** UPHELD. Exact calibration is at $r=0$; order is at $r<0$. CP2 must provide $r(\tau)$ and total-energy accounting.
- Objection: the earlier pure Gaussian vacuum is recovered.** UPHELD. The interacting restriction is generally non-Gaussian; the fourth-Hermite component gives an exact obstruction to a factorized free Gaussian eigenstate.

5. **Objection: CI8 or the PA-H1 three-mode space is an exact nonlinear invariant sector.** UPHELD . The local cubic force produces higher harmonics. Minimal-cutoff closure is a projection artifact.
6. **Objection: finite-volume ordering is quantum spontaneous symmetry breaking.** UPHELD . The quantum ground state is unique at finite cutoff and volume. No thermodynamic limit or SSB theorem is claimed.
7. **Objection: the node Hessian supplies a common Lorentz cone.** UPHELD . It is anisotropic by $21/2$, and the ultraviolet group speed is unbounded. A causal completion or replacement is required.
8. **Objection: negative relative energy means below empty space.** UPHELD . The subtraction is the critical parent ground energy, not an independently defined empty-space or no-condensate state.
9. **Objection: CP1 or Pre-A is closed.** UPHELD . CP1a supplies a common finite object and static reductions only. Dynamic crossing, total work, regulator removal, causal UV, evidence freeze, and holdout testing remain.

12. Exact status and next gate

The truthful verdict is:

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ADVANCE PA-CP1A-T3-CUBIC-SOS-COMMON-PARENT-v0
AS A STRUCTURAL COMPATIBILITY BENCHMARK ONLY.
DO NOT SELECT IT AS THE PHYSICAL EQUATION.
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It proves:

1. one finite T3 phase space, Weyl algebra, Hamiltonian family, volume, periodic boundary, $\hbar$, regulator, state-selection rule, and relative energy convention;
2. an exact $r=0$ PA-H1 three-mode quadratic calibration;
3. an exact $-9 \leq r < 0$ finite-regulator classical eight-node ordering theorem;
4. nontrivial local quartic coupling between the two roles; and
5. exact obstructions to unchanged componentwise-kernel matching, one-fixed-r composition, exact Gaussian state recovery, nonlinear three-mode closure, isotropic node propagation, and bounded ultraviolet speed.

It does not prove a physical vacuum, energy below empty space, cosmic cooling, a quantum or thermodynamic phase transition, BCC or any required morphology, spacetime, gravity, CP1, CP2, Pre-A, or Sector A.

The next decision is two-pronged:

1. CP1b/CP2: derive a nonstationary $r(\tau)$ crossing from a declared dynamical control pair or interacting state, with total work included; and
2. T-053: preregister the isotropy and ultraviolet-causality thresholds and reject this fitted fourth-order symbol unless a higher-invariant or causal completion passes them without refitting the same observables.

13. Reproduction and falsifier

Primary reproduction:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/pre_a_cp1a_t3_cubic_sos_common_parent.py
```

Non-importing independent reproduction:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/pre_a_cp1a_t3_cubic_sos_common_parent_independent.py
```

Integrated release verification:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/pre_a_cp1a_t3_cubic_sos_common_parent_verify.py
```

Falsifiers include failure of the derived coefficients, the exact lattice normal form or gap, the eight-node zero set, the node Hessian spectrum, the isometric symplectic calibration embedding, the $r=0$ -only ratio, either higher-harmonic coefficient, the finite-regulator minimizer proof, either stationary bound, the mixed quartic coefficient, the fourth-Hermite obstruction, the common-reference variational inequality, the anisotropic or unbounded-speed boundary, any pinned hash or verification count, or any claim that exceeds the explicit T0 scope above.

14. Prior-art and novelty boundary

Weyl CCR algebras over finite-dimensional phase spaces, confining Schrodinger ground-state uniqueness, sum-of-squares kernels, Fourier-mode variational trials, and Rayleigh--Ritz energy comparison are standard mathematics. No new general theorem in those subjects is claimed. The repository-specific contribution is the exact convention-level CP1a construction, the simultaneous calibration/ordering compatibility audit, the common energy ledger, and the explicit rejection criteria for using it as a physical theory.